

## Session 2

# An Intertemporal Theory of the Current Account

A euro-area reading of Schmitt-Grohe, Uribe and Woodford, chapters 3 and 5

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International Macroeconomics of the Euro Area | 3 hours | CY Cergy Paris Université

# How the three hours are built

*Two chapters, two exercises, two long breaks*

|                                                                                     |             |                                                    |        |
|-------------------------------------------------------------------------------------|-------------|----------------------------------------------------|--------|
|    | <b>0:00</b> | Where we left off                                  | 6 min  |
|    | <b>0:06</b> | Block 1 The household's problem                    | 40 min |
|    | <b>0:46</b> | Break                                              | 10 min |
|    | <b>0:56</b> | Block 2 Shocks and the current account, Exercise A | 35 min |
|    | <b>1:31</b> | Break                                              | 10 min |
|    | <b>1:41</b> | Block 3 Firms enter the model                      | 35 min |
|  | <b>2:16</b> | Stretch                                            | 5 min  |
|  | <b>2:21</b> | Block 4 Saving against investment, Exercise B      | 35 min |
|  | <b>2:56</b> | Wrap and problem set 2                             | 4 min  |

# Where we left off, and what is missing

*Last session ended on an identity. An identity explains nothing.*

$$CA = S - I$$

## What we proved

This holds in every country, in every year, by construction. It is accounting.

## What we never did

We never said what makes anyone save or invest. So we cannot yet say why a country runs a deficit.

## Today

Chapter 3 gives households a reason to borrow. Chapter 5 gives firms a reason to invest.

*The question the whole session answers: a country runs a current account deficit. Is that a problem, a symptom, or a sign that something is going right?*

# 1

## The household's problem

Chapter 3: why anyone would want to run a deficit

*40 minutes*

# A small open economy

*Two words that are doing a lot of work, especially inside a monetary union*

## Small

World prices and the world interest rate do not depend on anything this country does.

## Open

It trades goods and financial assets with the rest of the world.

## Who is small?

**Small and open:** the Netherlands, Austria, Portugal, Greece, Estonia, Latvia, and most of the euro area.

**Large and open:** Germany, the United States, Japan, the United Kingdom.

**Note where this is going: inside the euro area almost every member is small and takes the interest rate as given. In session 5 that stops being an assumption and becomes the defining constraint of the union.**

# The model economy

*Everything you need is on this slide*

## Two periods

Period 1 and period 2. Then the world ends.

## An endowment

Households receive  $Q_1$  goods in period 1 and  $Q_2$  in period 2. Goods are perishable, so you cannot store them.

## An inherited position

$B_0$ , carried in from the past, paying interest  $r_0$ .

## One choice

In period 1 the household picks consumption  $C_1$  and bond holdings  $B_1$ , which pay  $r_1$  in period 2.

**Because goods are perishable, the only way to move consumption between periods is to borrow or lend abroad. That is what makes this a theory of the current account.**

# From two budget constraints to one

*You have already met the step that does the work*

$$C1 + B1 - B0 = r0 B0 + Q1$$

Period 1: what you consume and save equals interest plus endowment

(1)

$$C2 + B2 - B1 = r1 B1 + Q2$$

Period 2: the same, one period on

(2)

$$B2 = 0$$

The transversality condition. Nobody is left to pay or collect.

(3)

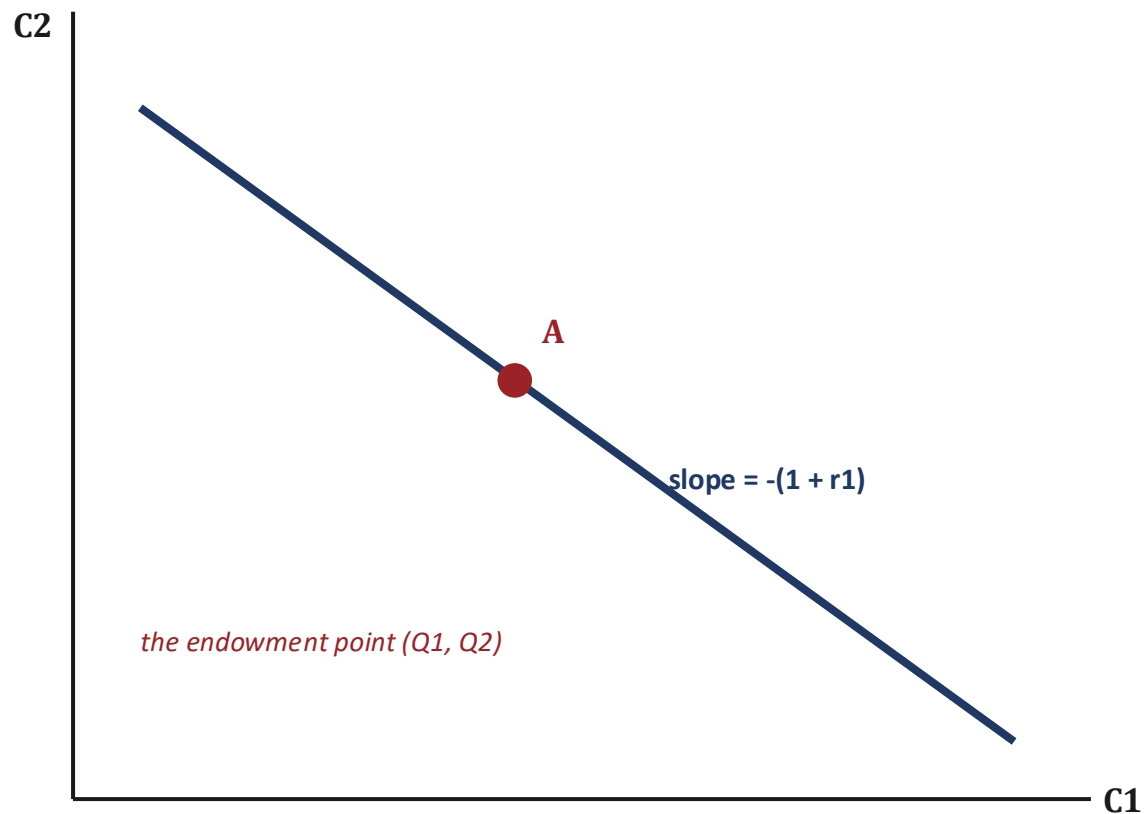
*Eliminate B1 and B2*

$$C1 + C2 / (1 + r1) = (1 + r0) B0 + Q1 + Q2 / (1 + r1)$$

*The present value of what you consume equals the present value of what you have. Note that  $B2 = 0$  is the same condition that drove the whole of chapter 2 last week.*

# The budget constraint is a line

*And the endowment point always sits on it*



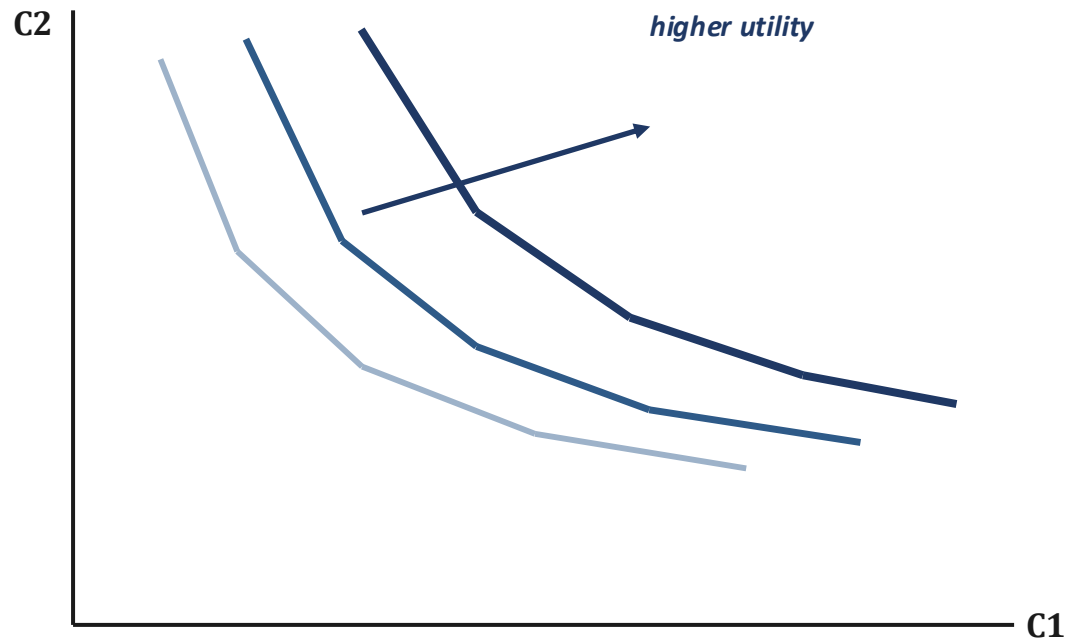
## Four things to notice

- It slopes down at minus one plus  $r$ . Give up a unit today, get  $1 + r$  units tomorrow.
- The endowment point  $A$  is always on the line. Consuming exactly your endowment is always affordable.
- Anything inside the triangle is feasible but wasteful: you died with money unspent.
- Anything outside is infeasible: you died owing money.

# What the household wants

*Lifetime utility, and the shape of indifference curves*

$$U(C1) + \text{beta } U(C2)$$



## Why they curve

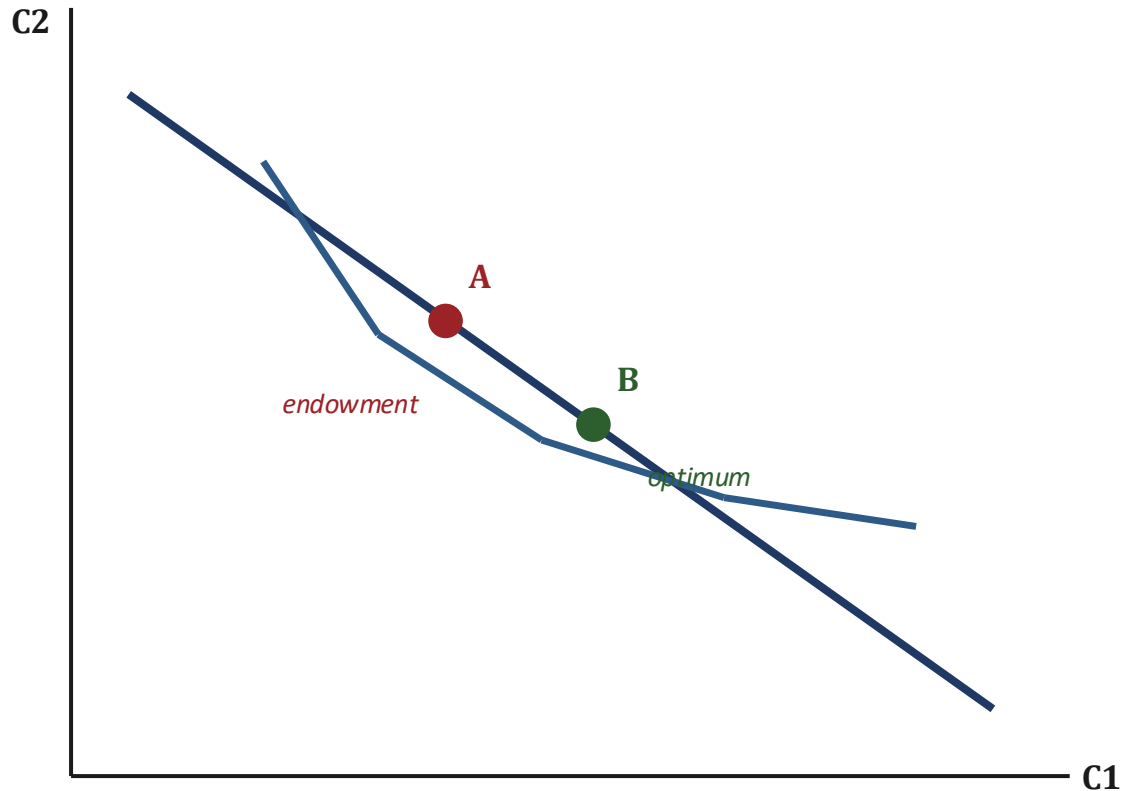
If you already consume a great deal today and almost nothing tomorrow, you will not give up much of tomorrow for one more unit today.

If you consume almost nothing today, you will give up a great deal of tomorrow for one more unit today.

That is diminishing marginal rate of substitution, and it is why people want to smooth.

# The optimum is a tangency

*And writing it down gives you the Euler equation*



*As drawn, B is to the right of A: the household borrows in period 1 and repays in period 2. It need not be. That depends on preferences, on the two endowments, and on the interest rate.*

$$U'(C1) = \text{beta} (1 + r1) U'(C2)$$

## The consumption Euler equation

At the optimum the slope of the indifference curve equals the slope of the budget line.

In words: the household is just indifferent between consuming one more unit today and saving it to consume  $1 + r$  units tomorrow.

If it were not indifferent, it would still be moving.

# Closing the model

*Free capital mobility pins down the interest rate*

$$r_1 = r^*$$

*The interest rate parity condition. If the domestic rate differed from the world rate, somebody could borrow in one market, lend in the other, and make unlimited profit.*

**The trade balance**

$$TB_1 = Q_1 - C_1$$

**The current account**

$$CA_1 = TB_1 + r_0 B_0$$

**So what determines a country's current account?**

Preferences, the two endowments, the world interest rate, and the position it inherited. That is the answer chapter 2 could not give. Everything in the next block is a comparative static on this list.

# Break, 10 minutes

When we come back we break the economy in three different ways  
and watch the current account respond.

# 2

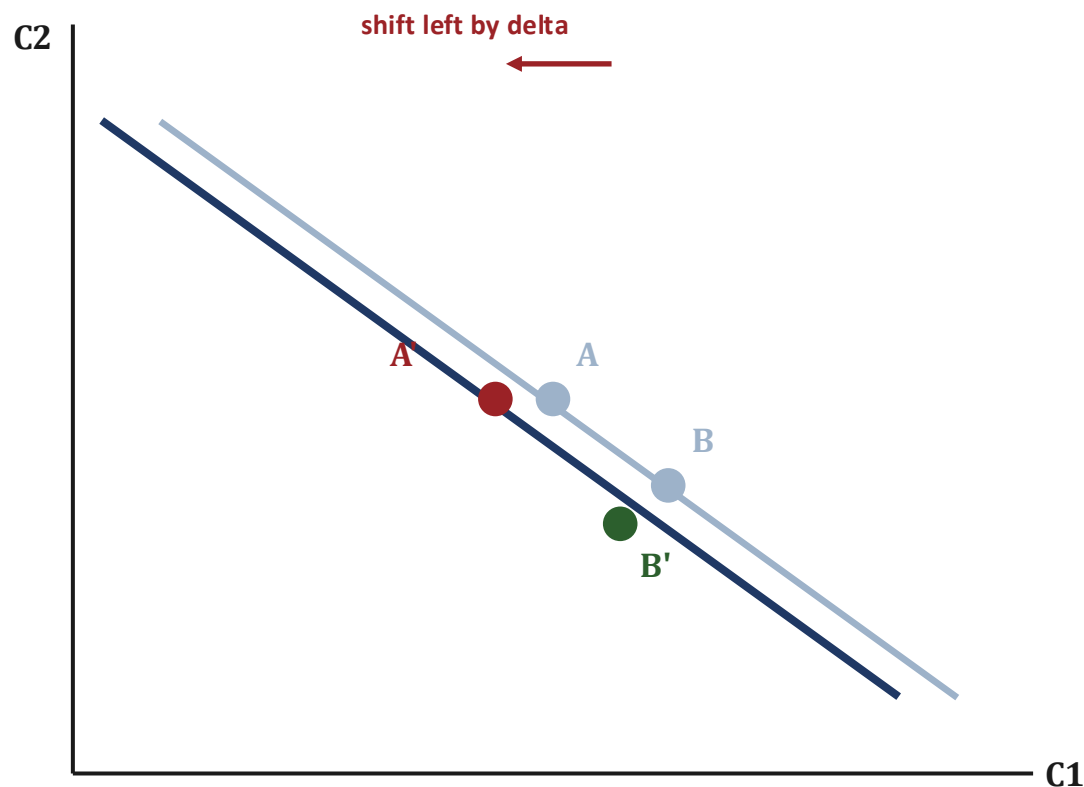
## Shocks and the current account

Chapter 3: temporary, permanent, and news about the future

*35 minutes*

# A temporary fall in output

*Q1 falls by delta. Q2 is unchanged.*



*In period 2 the country must run a larger surplus than it otherwise would, to repay the extra debt.*

## What the household does

- Both C1 and C2 are normal goods, so it wants to cut both.
- By cutting C2 as well, it does not have to cut C1 by the full delta.
- C1 falls by less than the shock.

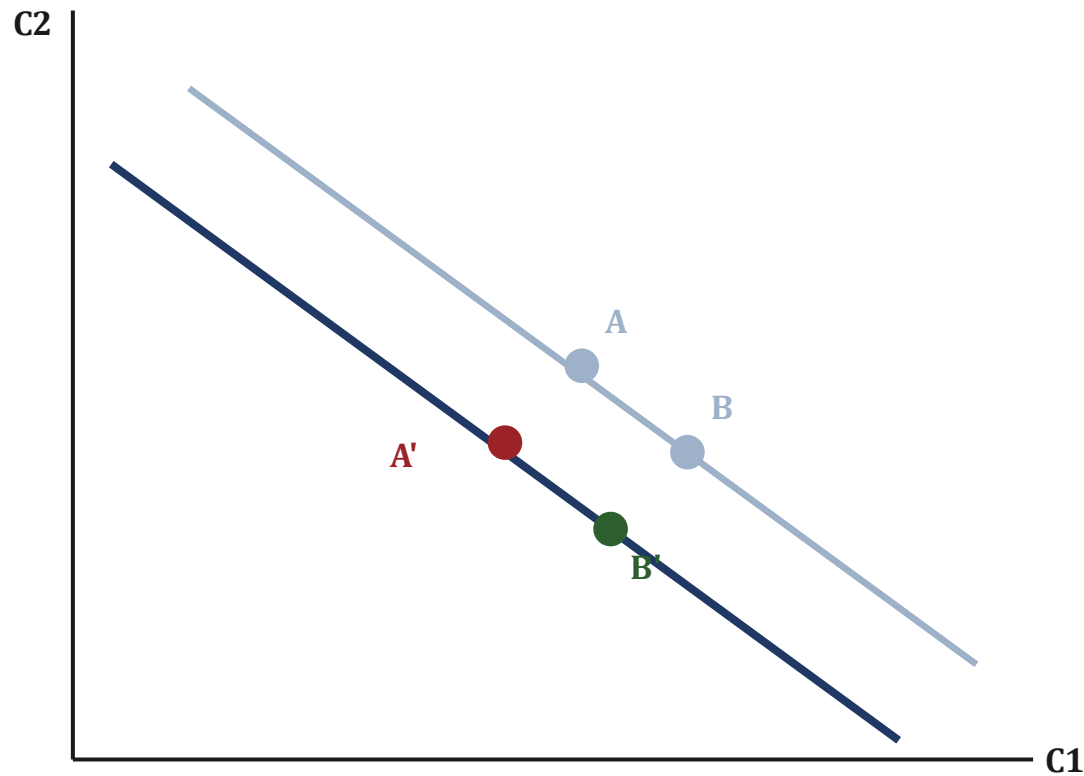
## What happens to the current account

Output fell by more than consumption, so the trade deficit widens and the country borrows abroad.

The current account deteriorates.

# A permanent fall in output

*Now Q1 and Q2 both fall by delta*



## What is different

- The whole budget line shifts down in parallel. Lifetime wealth really is lower.
- There is no good time to shift consumption to, because both periods are poorer.
- Consumption falls by close to the full delta in both periods.

## What happens to the current account

Output and consumption fall by almost the same amount, so the gap between them barely moves.

The current account is close to unchanged.

# The general principle

*One sentence that will come back in every remaining session*

**Economies finance temporary shocks and adjust to permanent ones.**

## Temporary shock

Borrow or lend abroad. Keep consumption roughly where it was.

**Large current account movement**

## Permanent shock

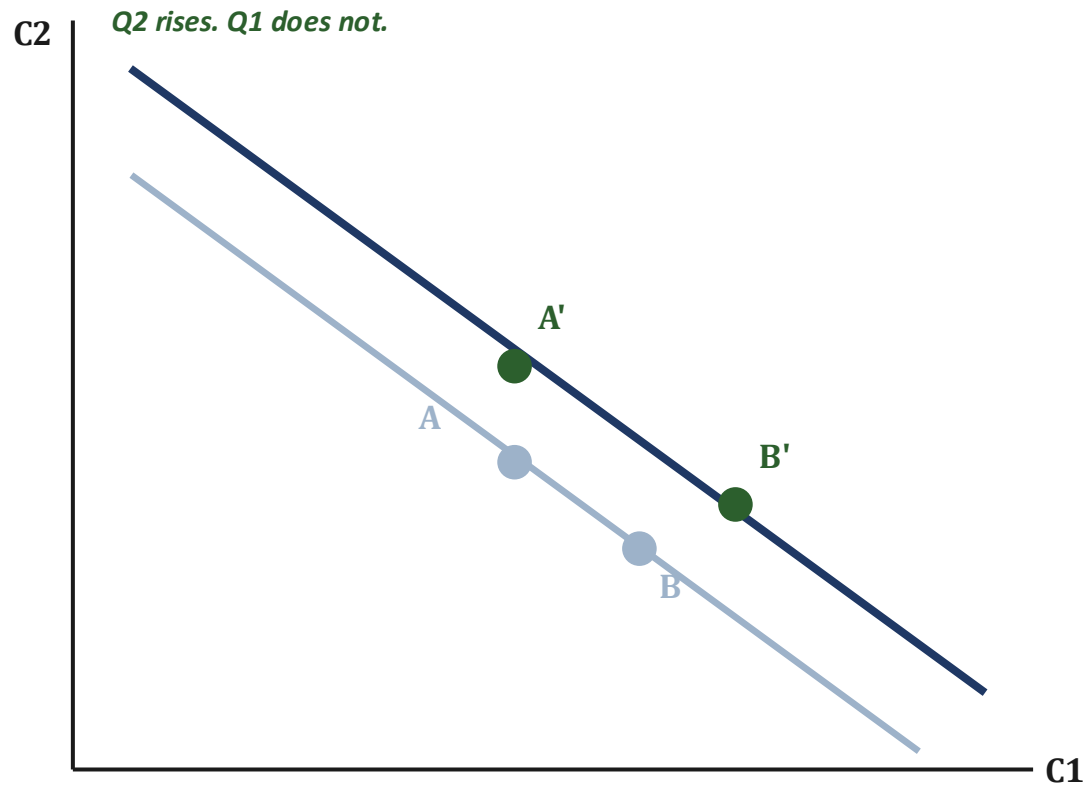
There is nothing to smooth against. Consumption moves with income.

**Little current account movement**

*This is also an identification strategy. We cannot observe what people expect. We can observe the current account, and it tells us.*

# Good news about the future

*The most important slide in this chapter for this course*



## The mechanism

- Lifetime wealth is higher, so consumption rises in both periods.
- But period-1 output has not changed.
- The extra consumption today is financed by borrowing from abroad.

**Good news about the future makes the current account worse.**

So a current account deficit is not necessarily a sign of a weak economy. It can be a sign that people expect to be richer later.

# Putting numbers on it

*Log preferences, no discounting. For the algebra track.*

$$C1 = \text{half of lifetime wealth}$$

$$C1 = (1/2) [ (1 + r_0) B_0 + Q1 + Q2 / (1 + r^*) ]$$

## Temporary rise in $Q1$

Consumption takes half, saving takes half.

**CA improves by half the shock**

## Permanent rise in both

Consumption takes nearly all of it.

**CA barely moves**

## Anticipated rise in $Q2$

Consumption rises today on income that has not arrived.

**CA deteriorates**

*The one-half comes from log utility with no discounting. Different preferences change the fraction, not the direction.*

# Exercise A: three shocks, three diagrams

*Fifteen minutes, pen and paper, in pairs*

## Draw it

- 1 Axes, budget line, endowment point A, optimum B. Mark the period-1 trade balance.
- 2 Shock 1: a two-year drought cuts this year's harvest. Draw the new A and B.
- 3 Shock 2: a permanent loss of an export market. Draw it again.
- 4 Shock 3: a large gas field is found, to be producing in six years. Draw it again.
- 5 For each, write down whether the current account improves or deteriorates, and why.

## Then argue about it

Two of these three produce a current account deficit, and they are not the two most people guess.

Swap papers with the pair next to you. If you disagree about shock 3, you have found the point of the chapter.

Extension: redo shock 1 assuming the drought is expected to last a decade. What changes, and why?

# Break, 10 minutes

So far nobody has built anything.  
After the break, firms arrive.

# 3

## Firms enter the model

Chapter 5: output is produced, not given

*35 minutes*

# One change to the model

*Until now output fell from the sky. Now somebody builds it.*

Chapter 3, this morning

Output was an endowment. Q just arrived.

Chapter 5, now

Output is produced with capital, and capital must be paid for first.

$$Q(t) = A(t) F(I(t-1))$$

**A(t)**

The level of technology, or productivity, in period t

**I(t-1)**

Investment made last period, which becomes productive this period

**The timing**

What you invest today determines what you can produce tomorrow. That delay drives the whole chapter.

# Three assumptions about $F$ , and one consequence

*No calculus needed. Each is a sentence about the real world.*

$$F(0) = 0$$

No capital, no output

You cannot produce without machines, buildings or equipment.

$$F' > 0$$

More capital, more output

Each extra machine adds something.

$$F'' < 0$$

But less each time

The tenth machine in a workshop adds less than the second.

**Marginal product of capital:  $MPK = A(t) F'(I(t-1))$**

The third assumption is the one that does the work. It makes the MPK schedule slope downward, and a downward-sloping schedule is what gives the firm a finite answer to how much to invest.

# What the firm does

*Borrow in period 1, produce in period 2, repay with interest*

**Period 1**

The firm borrows to buy capital goods.

$$D(1) = I(1)$$

**Period 2**

It produces, sells the output, repays the loan with interest.

$$\text{Profit} = A(2) F(I(1)) - (1 + r(1)) I(1)$$

**The choice**

Pick  $I(1)$  to make that profit as large as possible.

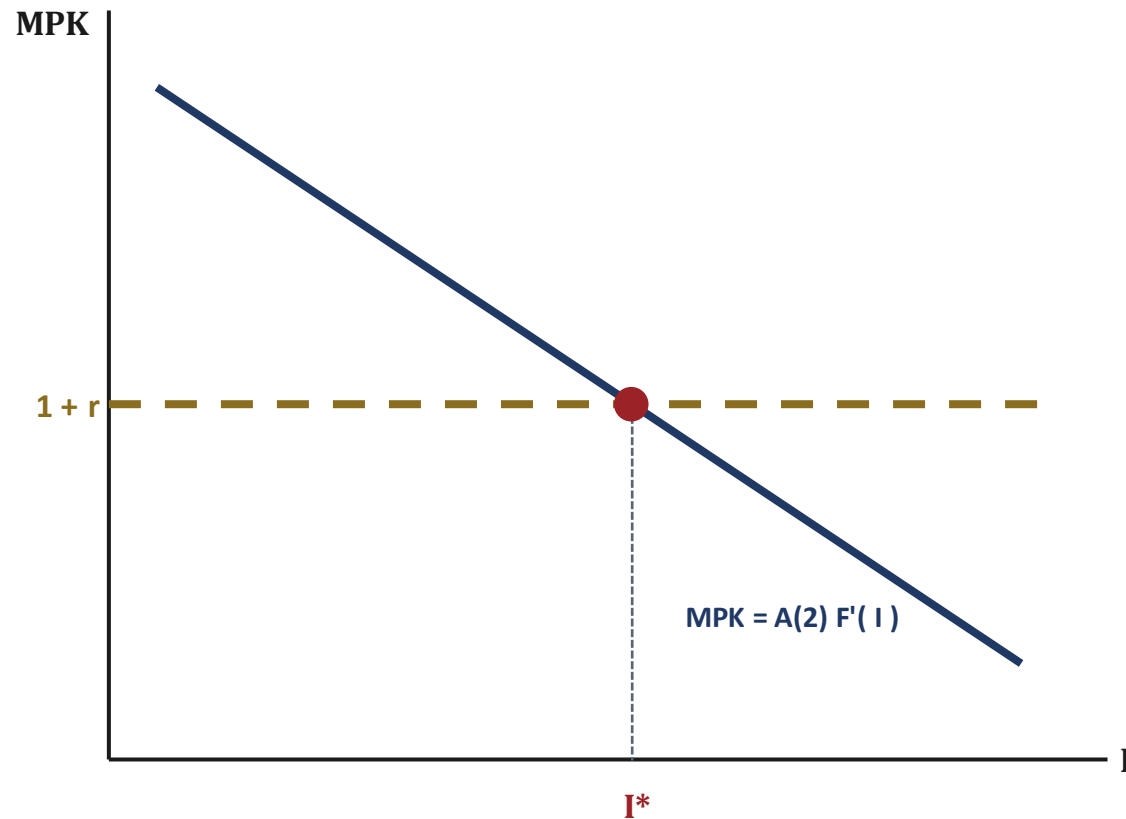
**max over  $I(1)$**

$$A(2) F'(I(1)) = 1 + r(1)$$

*Invest up to the point where the marginal product of capital equals the marginal cost of capital. That is the whole investment theory.*

# The same thing as a picture

*Why the firm stops investing where it does*



## Reading the diagram

- The downward line is what the next machine earns. It falls because of the third assumption.
- The flat line is what the next machine costs: the interest rate, the same at any scale.
- Left of  $I^*$ , the machine earns more than it costs, so buy it.
- Right of  $I^*$ , it costs more than it earns, so do not.
- Profit is the triangle between the two lines.

# Two things move investment

*And only two, in this model*

## The interest rate rises

The flat cost line shifts up and cuts the MPK schedule further left.

**Investment falls**

## Expected productivity rises

The MPK schedule shifts up and right and cuts the cost line further right.

**Investment rises**

**Hold on to both of these. In the last block, euro membership delivers them to the periphery at the same time.**

# Two schedules, and why they move in opposite directions

*Aggregate the firms, then aggregate the households*

$$I = I(r; A2)$$

$$S = S(r; A1, A2)$$

|                             |                                                                                       |                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>r rises</b>              | Investment falls. Capital costs more.                                                 | Saving rises. Postponing consumption pays better.                           |
| <b>A1 rises, today only</b> | No effect. Investment pays off tomorrow, so today's productivity is irrelevant to it. | Saving rises. Income is up today, and half of a temporary gain is saved.    |
| <b>A2 rises, expected</b>   | Investment rises. Tomorrow's capital is more productive.                              | Saving falls. Households consume today against income that has not arrived. |

Look at the bottom row. Expected future productivity pushes investment up and saving down at the same time. Both widen the deficit.

# Stretch, 5 minutes

Last block. We put the two schedules on one diagram  
and then let Spain and Ireland into the room.

# 4

## Saving against investment

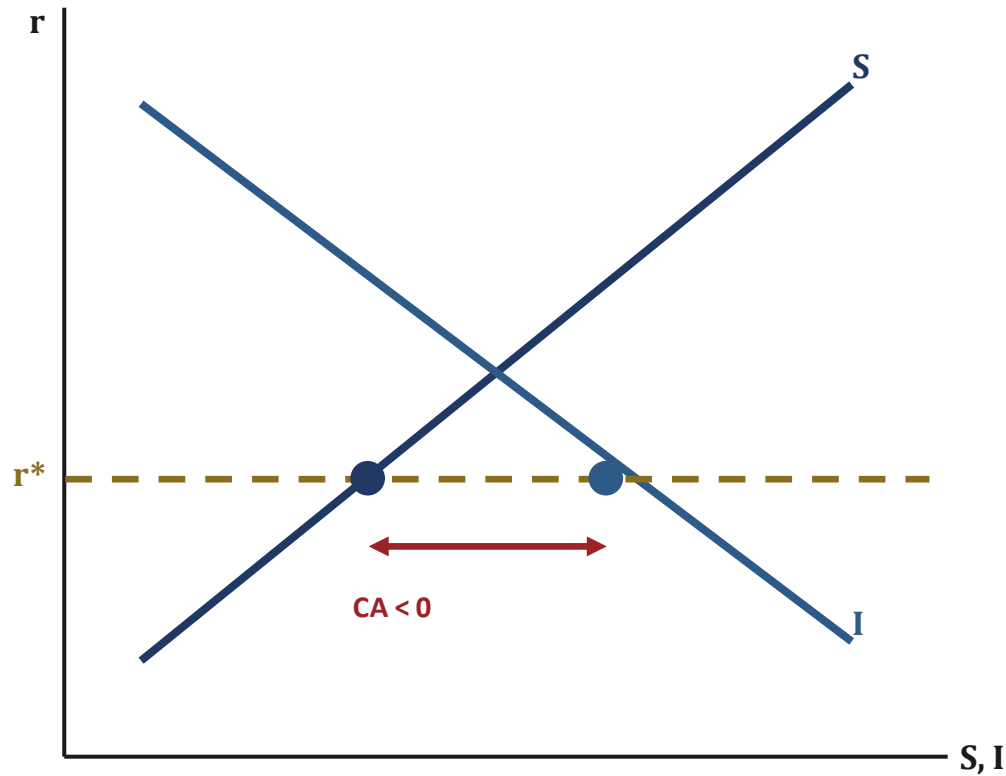
The Metzler diagram, and what the euro did to the periphery

*35 minutes*

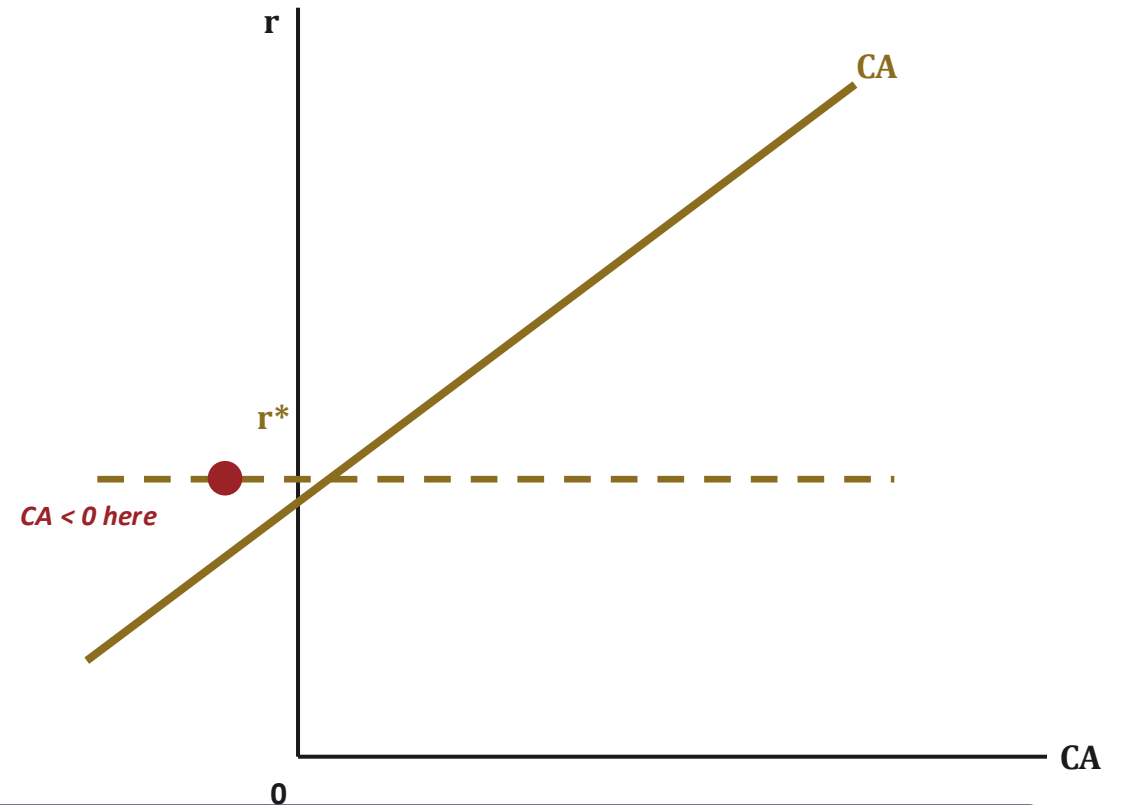
# The Metzler diagram

*The current account is the horizontal gap between the two schedules*

(a) Saving and investment



(b) The current account schedule



Panel (b) is just the horizontal distance between the two lines in panel (a), plotted against the interest rate.

# Three shocks, in an open economy

*The whole of chapter 5 in one table*

|                 | World rate $r^*$ rises | A1 rises (today) | A2 rises (expected) |
|-----------------|------------------------|------------------|---------------------|
| Saving          | up                     | up               | down                |
| Investment      | down                   | no change        | up                  |
| Current account | up                     | up               | down                |

## The column that matters for Europe

Look at the third column. Expected future productivity is the only shock that moves saving and investment in opposite directions, so it is the one that moves the current account most. An economy that believes it is about to get richer will borrow heavily, invest heavily, and run a large deficit. Nothing has gone wrong.

# The euro, read as two shocks at once

*What happened to the periphery after 1999*

## **$r^*$ falls, sharply**

Before the euro, Spain, Portugal, Greece and Ireland borrowed at high rates with a devaluation risk priced in. Monetary union removed that premium almost overnight.

**Investment up, saving down**

## **$A^2$ rises, or seems to**

Convergence to German income levels was widely expected. Capital would flow to where it was scarce and returns were high.

**Investment up, saving down**

**Both shocks push the same way. The model says the periphery should have run large, investment-driven current account deficits. It did. On the model's own terms, that was the system working.**

# Exercise B: was it investment or was it saving?

*Twenty minutes, Eurostat, in pairs*

1

Open Eurostat and pull gross fixed capital formation and gross national saving as shares of GDP, 1999 to 2010.

2

Do it for Spain and Ireland, plus your own country from session 1.

3

Chart them together. The vertical gap between the two is the current account.

4

Decide: did the deficit open up because investment rose, because saving fell, or both?

5

Mark 1999 and 2008 on your chart. What happens between them, and what happens after?

## What you are testing

The model made a specific prediction for the third column of the table: investment up, saving down, deficit wide.

If you find the deficit was driven mostly by investment, the model survives.

If you find it was driven by collapsing saving, the model is in trouble and you should say so.

Extension: add Germany on the same axes.

# The awkward question to end on

*The model explains the deficits. Does it excuse them?*

## What the model got right

The deficits were not anomalies. Capital flowed from where it was abundant to where it was scarce, exactly as the theory says it should.

## What the model assumes

That A2 actually arrives. The whole argument rests on the future productivity being real rather than merely expected.

## What happened next

In parts of the periphery the investment went into housing and non-tradables, which does not generate the export earnings needed to repay a foreign debt.

**So: was a ten per cent deficit a sign of success or a warning? On the evidence available in 2005, reasonable economists said success. Session 3 asks what the adjustment looks like when you are wrong and you have no exchange rate.**

# Before you go



## Problem set 2

Graphical questions on the three shocks, in both the endowment and the production economy. Starred: derive the Euler equation, and derive the one-half rule under log preferences.



## Reading

Schmitt-Grohe, Uribe and Woodford, chapters 3 and 5. Chapter 4 on the terms of trade is optional; the result is that the terms of trade multiply productivity everywhere.



## Keep your charts

The saving and investment series from exercise B come back in session 3, when we ask how the periphery closed those deficits.



## Next session

Chapters 9 and 10: the real exchange rate inside a monetary union. What adjustment looks like when there is no currency to devalue.